

TE Curve Verification Pipeline Familywise ONNX Report - wave3_2_clean_sequential_residual_offset

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_clean_sequential_residual_offset/2026-07-11-10-22-05_track2_wave3_2_clean_sequential_residual_offset_familywise_onnx_report ;`
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_clean_sequential_residual_offset/2026-07-11-10-22-05_track2_wave3_2_clean_sequential_residual_offset_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml ;`
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_clean_sequential_residual_offset/2026-07-11-10-22-05_track2_wave3_2_clean_sequential_residual_offset_familywise_onnx_report/model_inventory.csv ;`
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_2_clean_sequential_residual_offset/2026-07-11-10-22-05_track2_wave3_2_clean_sequential_residual_offset_familywise_onnx_report/per_curve_metrics.csv .`

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave3_2_clean_sequential_residual_offset` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave3_2_clean_sequential_residual_offset_fw_simplified_setpoints</code>	<code>2026-07-10-12-54-53_te_wave3_2_clean_sequential_residual_offset_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave3_2_clean_sequential_residual_offset_bw_simplified_setpoints</code>	<code>2026-07-10-13-09-01_te_wave3_2_clean_sequential_residual_offset_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave3_2_clean_sequential_residual_offset_global_simplified_setpoints</code>	<code>2026-07-10-12-38-46_te_wave3_2_clean_sequential_residual_offset_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/forward/2026-07-10-12-54-53_te_wave3_2_clean_sequential_residual_offset_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/forward/2026-07-10-12-54-53_te_wave3_2_clean_sequential_residual_offset_fw_simplified_setpoints/python/sequential_residual_offset_probe-epoch=074-val_mae=0.00371625.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/backward/2026-07-10-13-09-01_te_wave3_2_clean_sequential_residual_offset_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/backward/2026-07-10-13-09-01_te_wave3_2_clean_sequential_residual_offset_bw_simplified_setpoints/python/sequential_residual_offset_probe-epoch=151-val_mae=0.00364800.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/global/2026-07-10-12-38-46_te_wave3_2_clean_sequential_residual_offset_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/global/2026-07-10-12-38-46_te_wave3_2_clean_sequential_residual_offset_global_simplified_setpoints/python/sequential_residual_offset_probe-epoch=110-val_mae=0.00370701.ckpt</code>

Aggregate Metrics

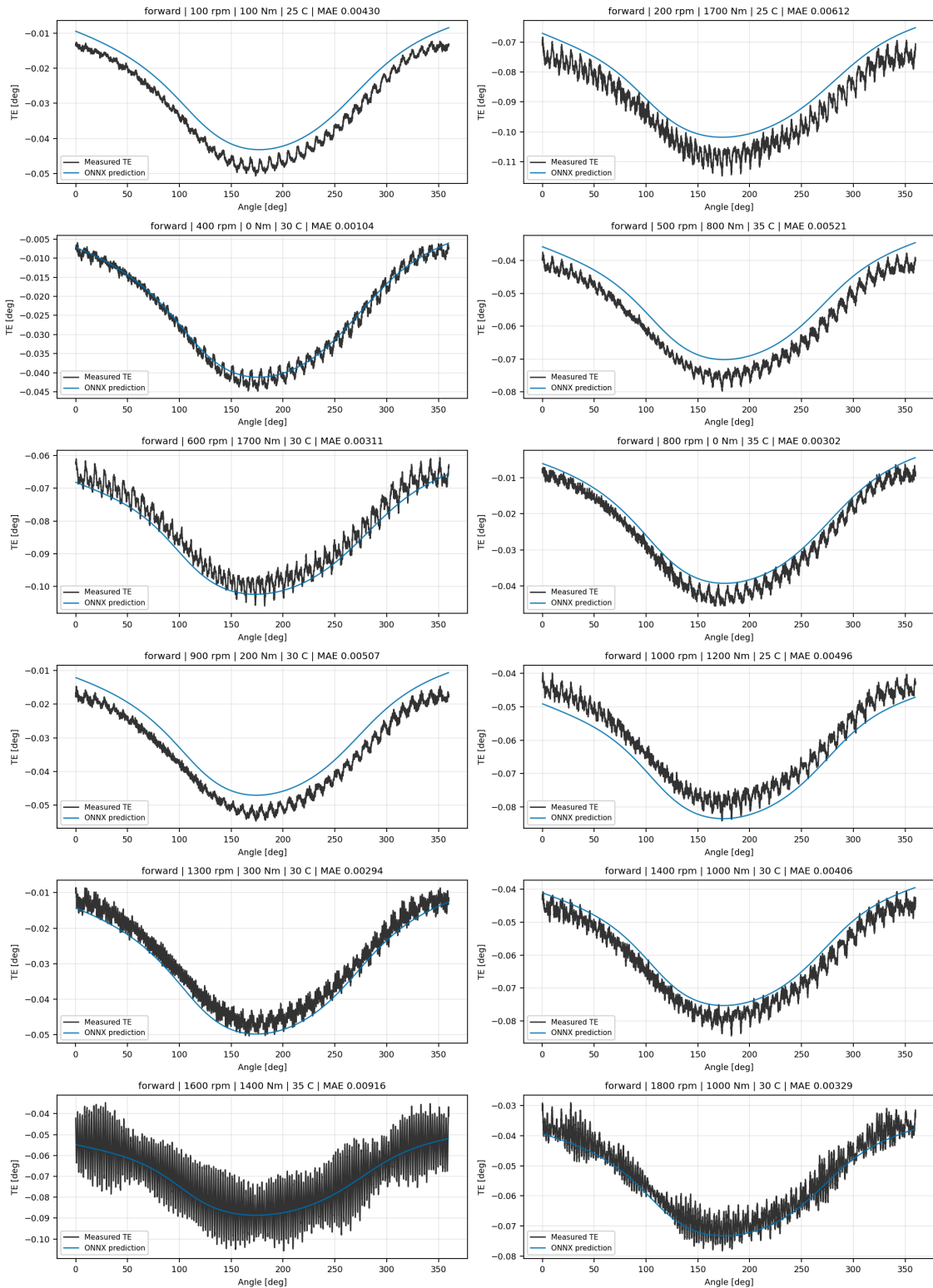
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003475	0.003912	8.132	13.778
backward	97	0.003577	0.004001	8.280	13.551
global	194	0.003636	0.004065	8.486	15.133

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000198	0.002938	0.001734	0.006182
backward	-0.000108	0.002850	0.001851	0.007030
global	0.000415	0.003059	0.001788	0.007024

Forward 12-Curve Page

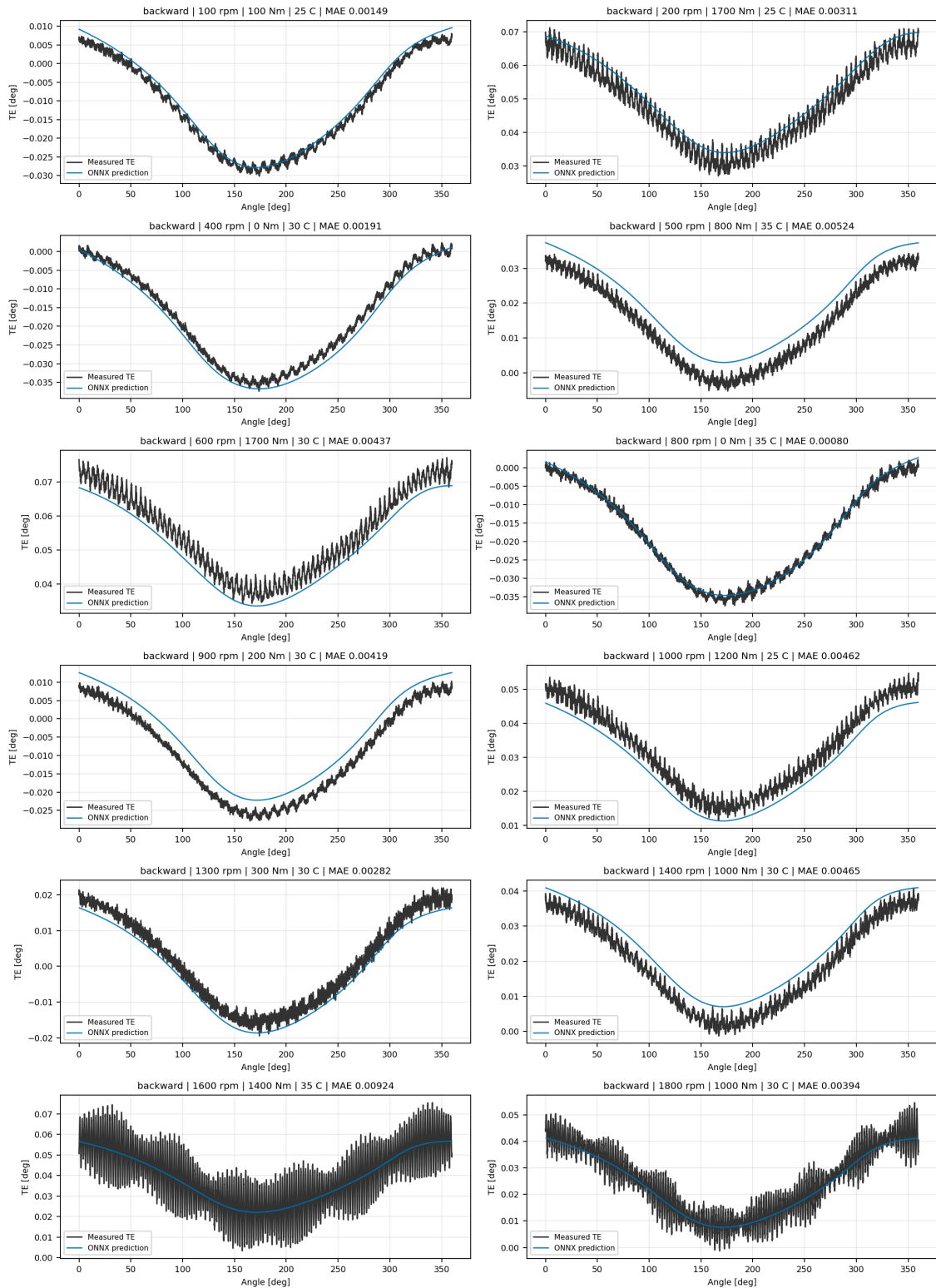
Simplified Dataset + Setpoints | wave3_2_clean_sequential_residual_offset | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

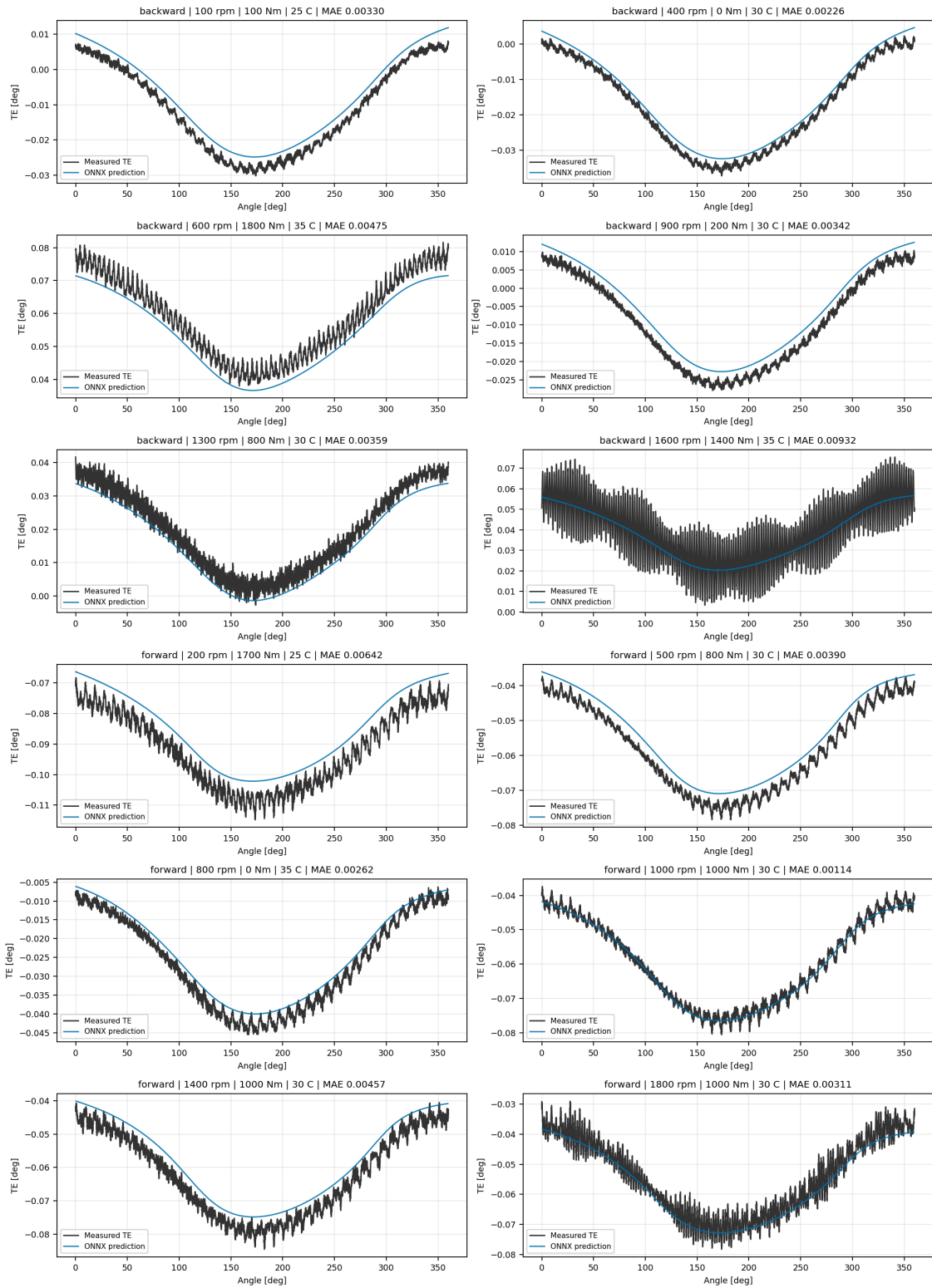
Simplified Dataset + Setpoints | wave3_2_clean_sequential_residual_offset | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave3_2_clean_sequential_residual_offset | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave3_2_clean_sequential_residual_offset ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_2_clean_sequential_residual_offset_fw_polished_setpoints	2026-07-10-14-12-44_te_wave3_2_clean_sequential_residual_offset_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave3_2_clean_sequential_residual_offset_bw_polished_setpoints	2026-07-10-14-35-10_te_wave3_2_clean_sequential_residual_offset_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave3_2_clean_sequential_residual_offset_global_polished_setpoints	2026-07-10-13-46-46_te_wave3_2_clean_sequential_residual_offset_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/forward/2026-07-10-14-12-44_te_wave3_2_clean_sequential_residual_offset_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/forward/2026-07-10-14-12-44_te_wave3_2_clean_sequential_residual_offset_fw_polished_setpoints/python/sequential_residual_offset_probe-epoch=092-val_mae=0.00219799.ckpt
backward	models/polished_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/backward/2026-07-10-14-35-10_te_wave3_2_clean_sequential_residual_offset_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/backward/2026-07-10-14-35-10_te_wave3_2_clean_sequential_residual_offset_bw_polished_setpoints/python/sequential_residual_offset_probe-epoch=089-val_mae=0.00218161.ckpt
global	models/polished_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/global/2026-07-10-13-46-46_te_wave3_2_clean_sequential_residual_offset_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_2_clean_sequential_residual_offset/global/2026-07-10-13-46-46_te_wave3_2_clean_sequential_residual_offset_global_polished_setpoints/python/sequential_residual_offset_probe-epoch=128-val_mae=0.00217355.ckpt

Aggregate Metrics

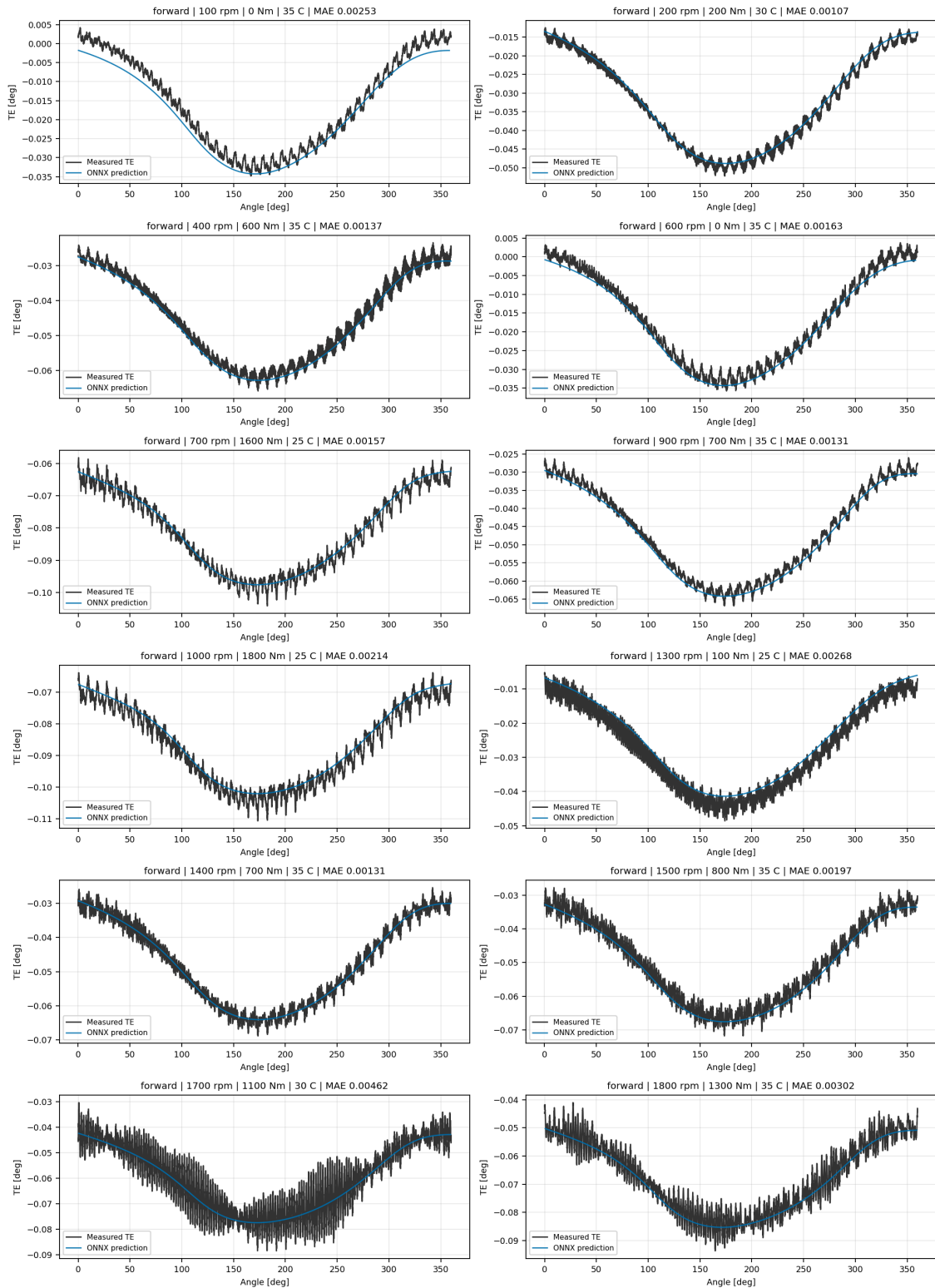
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002253	0.002715	5.000	10.589
backward	94	0.002729	0.003236	5.078	11.315
global	194	0.002454	0.002944	4.962	11.305

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000029	0.001055	0.001911	0.008038
backward	0.000213	0.001004	0.002372	0.008659
global	0.000157	0.000977	0.002126	0.009069

Forward 12-Curve Page

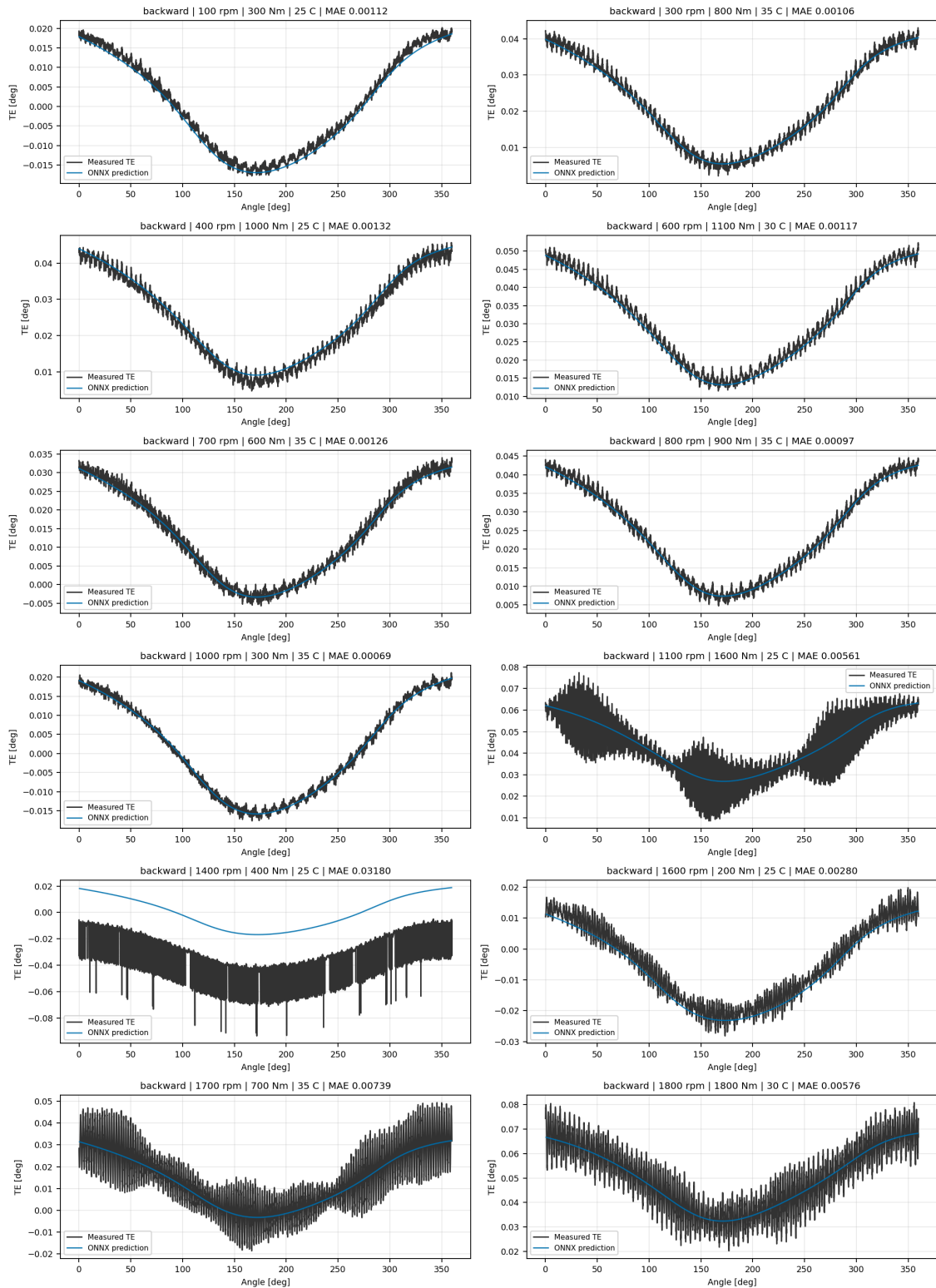
Polished Dataset + Setpoints | wave3_2_clean_sequential_residual_offset | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

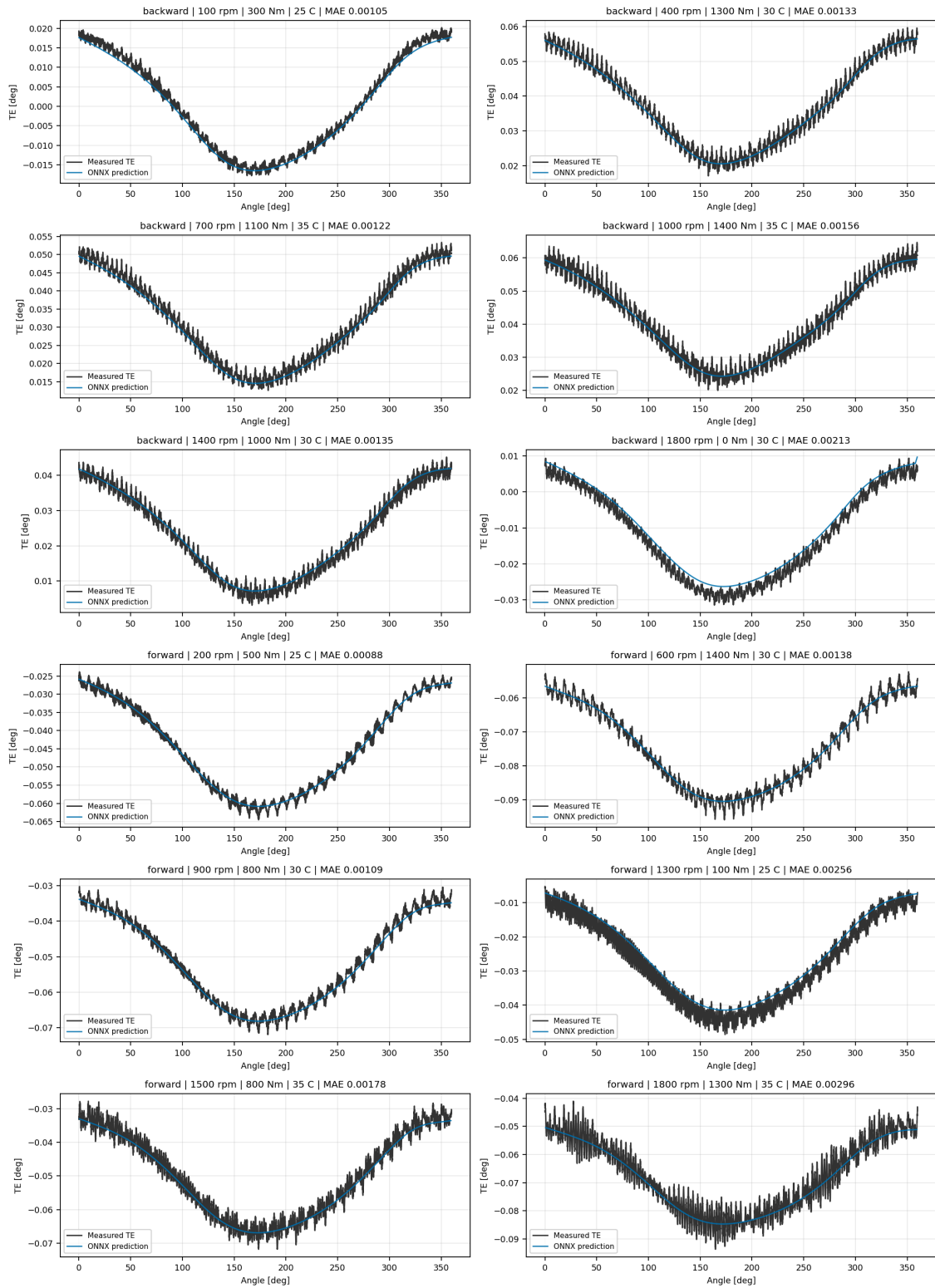
Polished Dataset + Setpoints | wave3_2_clean_sequential_residual_offset | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave3_2_clean_sequential_residual_offset | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave3_2_clean_sequential_residual_offset ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_2_clean_sequential_residual_offset_fw_polished_actual_values	2026-07-10-22-06-08_te_wave3_2_clean_sequential_residual_offset_fw_polished_actual_values	polished_point_v1
backward	te_wave3_2_clean_sequential_residual_offset_bw_polished_actual_values	2026-07-10-22-37-04_te_wave3_2_clean_sequential_residual_offset_bw_polished_actual_values	polished_point_v1
global	te_wave3_2_clean_sequential_residual_offset_global_polished_actual_values	2026-07-10-21-34-26_te_wave3_2_clean_sequential_residual_offset_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave3_2_clean_sequential_residual_offset/forward/2026-07-10-22-06-08_te_wave3_2_clean_sequential_residual_offset_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_2_clean_sequential_residual_offset/forward/2026-07-10-22-06-08_te_wave3_2_clean_sequential_residual_offset_fw_polished_actual_values/python/sequential_residual_offset_probe-epoch=153-val_mae=0.00216939.ckpt
backward	models/polished_dataset/actual_values/exported/wave3_2_clean_sequential_residual_offset/backward/2026-07-10-22-37-04_te_wave3_2_clean_sequential_residual_offset_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_2_clean_sequential_residual_offset/backward/2026-07-10-22-37-04_te_wave3_2_clean_sequential_residual_offset_bw_polished_actual_values/python/sequential_residual_offset_probe-epoch=127-val_mae=0.00219453.ckpt
global	models/polished_dataset/actual_values/exported/wave3_2_clean_sequential_residual_offset/global/2026-07-10-21-34-26_te_wave3_2_clean_sequential_residual_offset_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_2_clean_sequential_residual_offset/global/2026-07-10-21-34-26_te_wave3_2_clean_sequential_residual_offset_global_polished_actual_values/python/sequential_residual_offset_probe-epoch=151-val_mae=0.00216362.ckpt

Aggregate Metrics

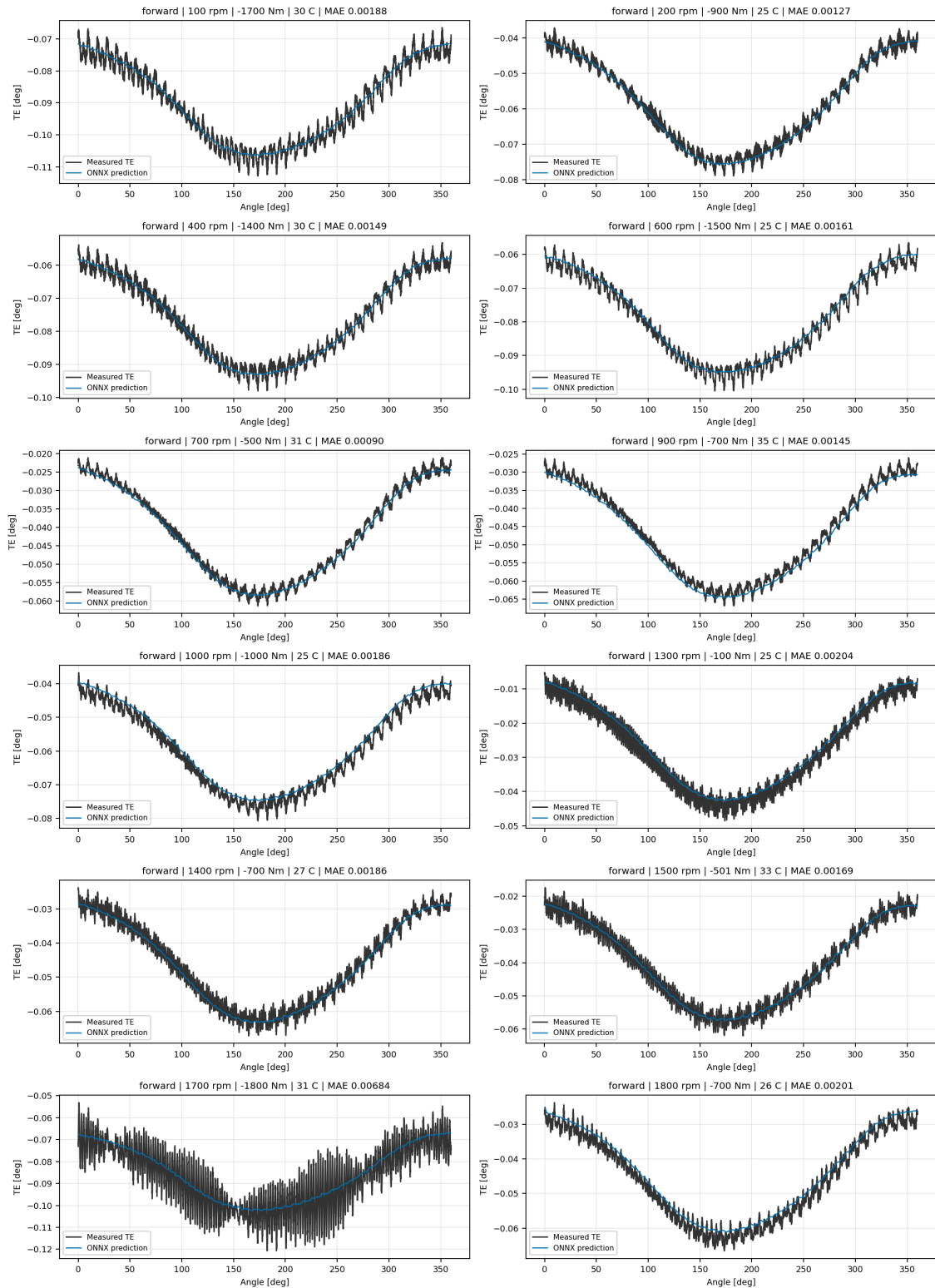
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002138	0.002589	4.710	9.504
backward	94	0.002472	0.002980	4.774	11.059
global	194	0.002302	0.002798	4.758	10.511

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000221	0.000841	0.001911	0.007860
backward	0.000031	0.000552	0.002377	0.008688
global	-0.000060	0.000777	0.002122	0.008373

Forward 12-Curve Page

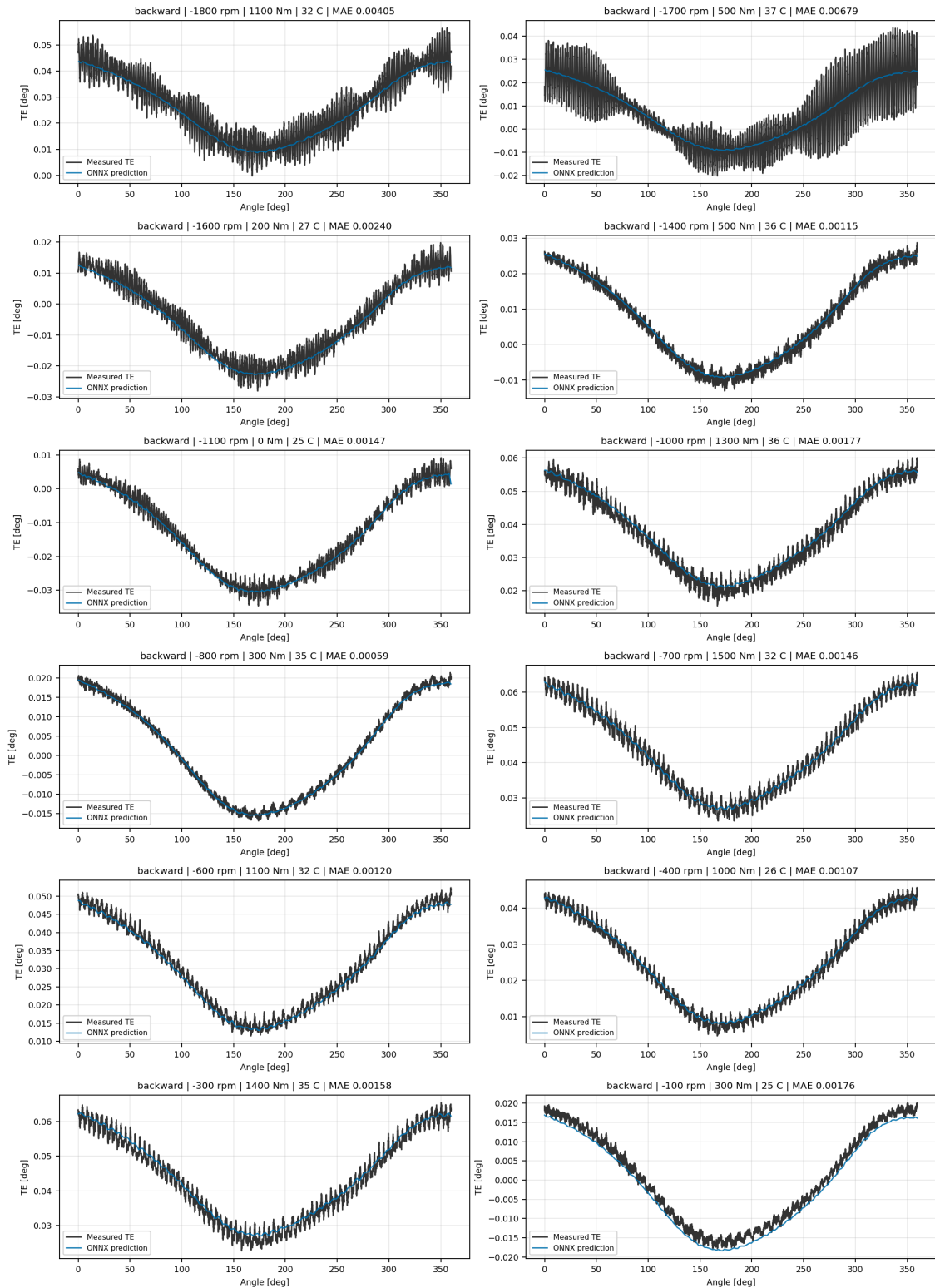
Polished Dataset + Actual Values | wave3_2_clean_sequential_residual_offset | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

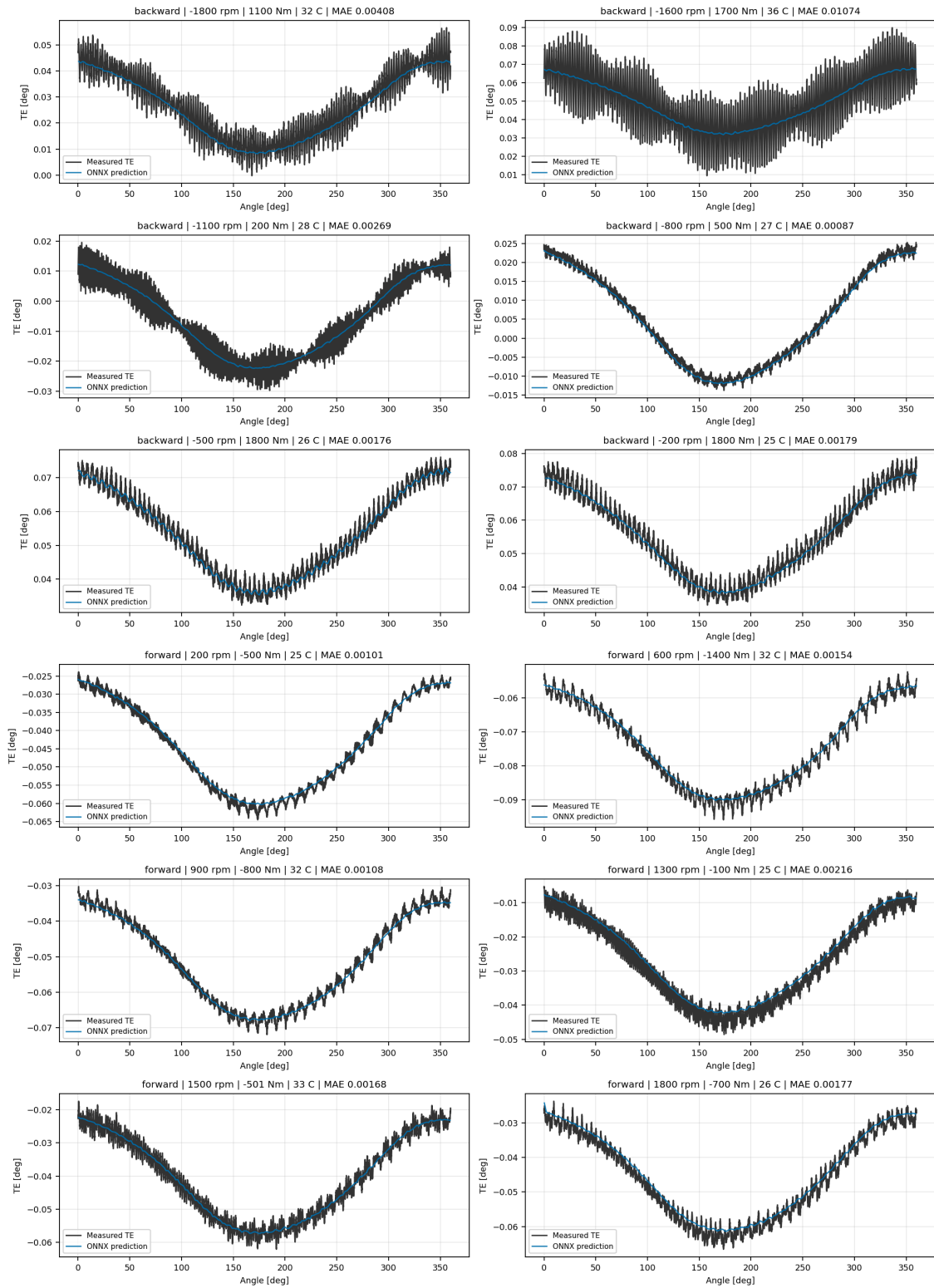
Polished Dataset + Actual Values | wave3_2_clean_sequential_residual_offset | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave3_2_clean_sequential_residual_offset | global



polished_dataset_actual_values global 12-curve collage